Www.Medicalstudyzone.com

GASTROINTESTINAL HORMONES

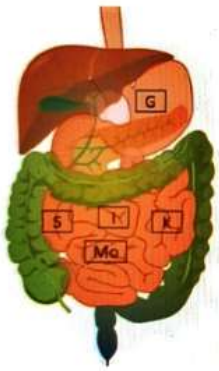
----- Active space --

GI hormones act upon the GI tract and the brain.

Classification of GI hormones : 3 main families.

Gastrin family	Secretin family	Others
1. Gastrin. 2. Cholecystokinin (CCK).	1. Secretin. 2. Glucagon. 3. Vasoactive Intestinal Peptide (VIP). 4. Gastric Inhibitory Peptide (GIP).	1. motilin. 2. Guanylin. 3. Peptide YY. 4. Ghrelin.

Enteroendocrine cells :



Cells	Hormones secreted
G-cells	Gastrin
I-cells	Cholecystokinin
S-cells	Secretin
K-cells	GIP
mo cells	motilin

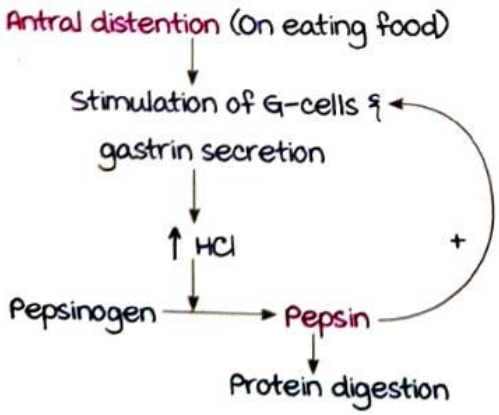
Note : M-cells are microfold cells of peyer's patches (In terminal ileum).

Gastrin family

00:05:23

GASTRIN :

Secretion :



Factors affecting release of gastrin.

Increase	Decrease
• Antral distention. • Protein-rich diet (↑ pepsin). • Gastrin-releasing peptide (GRP) by vagal nerve endings. • Ca^{2+}. • Epinephrine.	• Excess acid. • Somatostatin (Universal inhibitor). • GIP. • Secretin.

Www.Medicalstudyzone.com

----- Active space -----

Actions :

- Increases acid production via CCK-B receptor
- Lowers esophageal sphincter (LES) contraction → Prevents reflux (Protective)
- ↑ gastric motility
- ↑ pepsin levels
- Trophic action : Stimulates GI tract mucosal growth

CHOLECYSTOKININ – PANCREOZYMIN (CCK-PZ) :

Secretion and actions of CCK-PZ :

Intake of food rich in fats, proteins → Stimulation of I-cells —Release→ CCK-PZ.

	Actions	
	CCK	PZ
Specific actions	Via CCK-A receptor : • Cholagogue contraction of gall bladder. • Release of bile : Relaxation of Sphincter of Oddi.	Stimulates secretion of enzyme-rich pancreatic juice
GI actions	• Augments action of secretin. • Stimulates intestinal and colonic motility. • Inhibits gastric emptying.	
CNS actions	• ↓ food intake by ↑ satiety. • Implicated in anxiety. • Aids in analgesia.	

Secretin family

00:17:20

SECRETIN :

AKA natural antacid.

First hormone to be discovered (Bayliss & Starling).

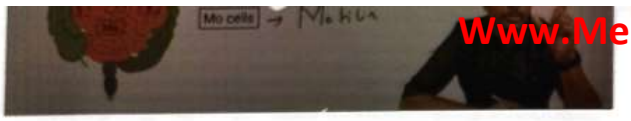
Secretion :

Acid-rich chyme → Duodenum —Stimulates→ S-cells —Secrete→ secretin.

Actions :

- Release of HCO_3^- rich (Alkaline) pancreatic juice : For acid neutralization.
- Augments action of CCK.
- Decreases gastric acid secretion.
- Contraction of pyloric sphincter : Prevents food from moving into intestine.

Feedback



Www.Medicalstudyzone.com

----- Active space -----

GASTRIC INHIBITORY POLYPEPTIDE (GIP) :

Now known as glucose dependent Insulinotropic polypeptide (GIP).

Secretion :

Oral intake of glucose $\xrightarrow{\text{Stimulates}}$ K-cells $\xrightarrow{\text{release}}$ GIP $\xrightarrow{\text{Stimulates}}$ β cells in pancreatic islets ⟶ ↑ insulin

GI hormones that ↑ insulin secretion : Incretins. Eg : GIP and Glucagon-like peptide (GLP).

VASOACTIVE INTESTINAL POLYPEPTIDE (VIP) :

Actions

Salivary glands	Stomach	Intestine
Potentiate action of acetylcholine	↓ acid production	• ↑ fluid and electrolyte secretions. • Relaxation of smooth muscle.

VIPoma :

Neuroendocrine tumor (↑ secretion of VIP).

Leads to : WDHA syndrome : AKA pancreatic cholera/verner - morrison syndrome.

- Watery diarrhea (D/t ↑ fluid/electrolyte secretion)
- Hypokalemia (↑ K^+ secretion)
- Achlorhydria.

Other Hormones 00:27:10

MOTILIN :

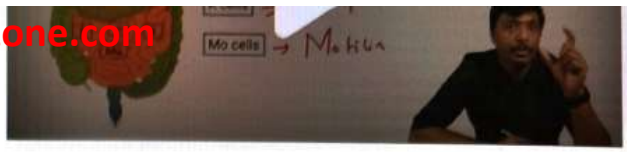
- Seen in all fasting individuals.
- Causes ↑ GI motility : Regulates migratory motor complex (mmc) from stomach to colon.
- Erythromycin : Agonist to motilin receptor ⟶ used to treat gastroparesis in diabetic patients.

GUANYLIN :

- ↑ Cl^- secretion in GI tract.
- Structurally similar to E-coli ⟶ molecular mimicry : Ecoli act through guanylin receptor.

Feedback

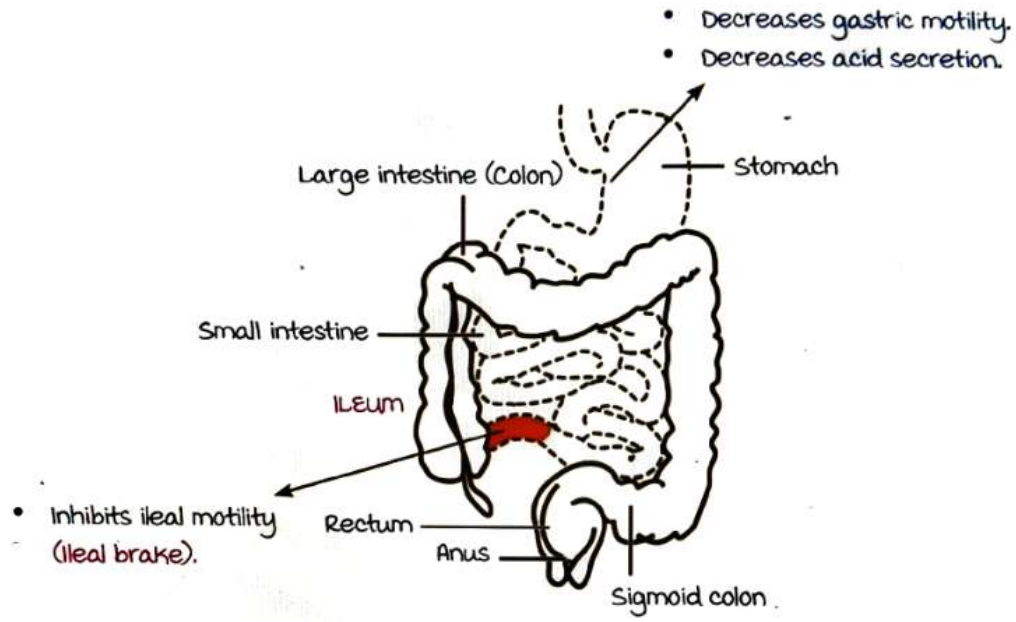
Www.Medicalstudyzone.com



----- Active space -----

PEPTIDE YY

Stimulus : Fat-rich diet.



GHRELIN

AKA Hunger hormone (Causes hunger contractions).

Secretion : From oxyntic gland in stomach.

- Levels peak during fasting.
- Levels are
  - High in anorexia nervosa patients.
  - Low in obese patients.

Actions :

1. Orexinergic : ↑ food intake.
2. Increases
   - Gastric acid.
   - Gastric motility.
3. ↑ adipogenesis.
4. ↑ release of Growth Hormone (GH) : GH-REL-IN.
   (Releasing)

Feedback